

Assignment # 1

ELIZA Mini-Lab — Exploring Weizenbaum’s “DOCTOR” (1966)

Course level: Senior undergrad / early graduate (intro NLP or AI history)

Time needed: ~20–40 minutes exploration + ~20 minutes reflection

No coding required

Why this activity?

Joseph Weizenbaum’s **ELIZA** (MIT, 1964–1966) is a rule-based conversation program. Its most famous script, **DOCTOR**, reflects user statements as questions using keyword-based **decomposition** and **reassembly** rules. This exercise lets you *experience* the “Eliza effect” first-hand and contrast it with modern LLMs.

Important: ELIZA is not a mental-health tool. If the topic is sensitive for you, look for professional help with a human psychotherapist. ITESO has a center called CJuven ITESO that can offer help for those who need it. Schedule an appointment at <https://cjuven.iteso.mx>

Learning objectives

By the end, you should be able to:

1. Explain, in two–three sentences, how ELIZA’s keyword rules create the **illusion** of understanding.
 2. Identify examples of the **Eliza effect** (anthropomorphizing a simple system).
 3. Contrast ELIZA’s pattern-matching with modern LLM behavior on the same prompts.
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Where to try ELIZA (online demos)

These load in your browser. Prefer demos that explicitly say they use the **1966 DOCTOR** script (or a faithful port).

1. **ELIZA Archaeology — “Try ELIZA” (with *trace mode)**
*Faithful reconstruction aiming to match Weizenbaum’s 1966 behavior. *trace shows which rules fired.*

<https://sites.google.com/view/elizaarchaeology/try-eliza>

2. **mass:werk ELIZA (elizabot.js) — classic JavaScript port**
Main demo, plus a terminal-style UI and a page that recreates the famous sample dialogue.

Main demo: <https://www.masswerk.at/elizabot/>
Terminal UI: <https://www.masswerk.at/elizabot/eliza.html>
“Eliza Test”: https://www.masswerk.at/elizabot/eliza_test.html

3. **mass:werk “E.L.I.Z.A.” — speech I/O variant**
Adds speech recognition/synthesis on top of the ELIZA port (same underlying rules).

<https://www.masswerk.at/eliza/>

4. **NJIT ELIZA demo (JS variant)**
Simple, quick to try.

<https://web.njit.edu/~ronkowitz/eliza.html>

5. **Cal State Fullerton ELIZA demo (JS variant)**
Widely used classroom demo.

<https://psych.fullerton.edu/mbirnbaum/psych101/eliza.htm>

6. **Feel free to search the internet and look for an Eliza to your liking.**
Some times links just die and are no longer available, but for sure you’ll find one.

Optional: Source & historical artifacts

Original MAD-SLIP ELIZA source (scan): https://archive.org/details/eliza_1966_mad_slip_src
ELIZA recreation (code): <https://github.com/anthay/ELIZA>

Mini-activities (pick any 3)

A. Recreate the CACM dialogue

Open mass:werk’s *Eliza Test* page and step through the classic “Men are all alike.” conversation. Identify two lines where a simple substitution creates a persuasive reply.

B. Prompt swap

Try each input on a demo and write the reply; guess what keyword/rule fired.

- “I am unhappy.”
- “My mother takes care of me.”
- “Because I am tired.”
- “You don’t listen to me.”

C. Ambiguity probe

Test pronouns and negation (e.g., “he/ she/ they,” “don’t/ never”). When does ELIZA mis-reflect your wording? Why?

D. Memory & quitting

Does ELIZA remember facts you stated earlier (e.g., your name)? Find a quit phrase (e.g., “bye”) and record the final message.

E. Trace mode (ELIZA Archaeology)

Enter `*trace` to enable tracing, then converse briefly. Paste one trace snippet and explain, in 3–4 sentences, how the decomposition/reassembly worked.

What to submit (suggested, be creative)

1. At least **2 pages** (bullet points are fine) containing:
 - **3–5 short transcript snippets** (copy/paste or screenshots) from different demos.
 - For **two** snippets, a guess at the **keyword** and **reassembly** rule that fired.
 - **Three observations** comparing ELIZA vs. a modern LLM (e.g., handling of context, reasoning, consistency).
 2. (Optional) One `*trace` screenshot with a short explanation.
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Quick background (for reference)

- ELIZA uses **pattern-matching** over surface forms—no semantic parsing or world modeling.
 - The **DOCTOR** script reflects language back to the user in a Rogerian style.
 - The striking human-likeness many users report is called the **Eliza effect**—projecting understanding onto systems that do not have it.
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Academic integrity & ethics

- Do not present ELIZA’s output as clinical advice.
- You may collaborate on ideas, but **write your own reflections** and capture your *own* transcripts.

Grading rubric

- **Transcript evidence (30%)**: clear, varied examples from at least two demos.
 - **Rule insight (25%)**: plausible identification of keywords/rules; explanation is concise and accurate.
 - **Comparative analysis (25%)**: thoughtful, concrete contrasts with modern LLMs.
 - **Clarity & presentation (10%)**: organized, readable, within length.
 - **Professional/ethical awareness (10%)**: acknowledges limitations and ethical considerations.
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Optional readings

- Weizenbaum, J. (1966). *ELIZA—A computer program for the study of natural language communication between man and machine*. **Communications of the ACM**, 9(1), 36–45.
 - Jurafsky, D., & Martin, J. H. (2008). *Speech and Language Processing* (3rd ed.). (See first page of chapter 2 on Eliza)
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Quick copy block (just the links)

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ELIZA Archaeology (trace): https://sites.google.com/view/elizaarchaeology/try-eliza
mass:werk ELIZABOT:      https://www.masswerk.at/elizabot/
Terminal UI:             https://www.masswerk.at/elizabot/eliza.html
Eliza Test (classic):    https://www.masswerk.at/elizabot/eliza\_test.html
mass:werk E.L.I.Z.A.:    https://www.masswerk.at/eliza/
NJIT ELIZA:              https://web.njit.edu/~ronkowit/eliza.html
CSUF ELIZA:              https://psych.fullerton.edu/mbirnbaum/psych101/eliza.htm
Original source scan:    https://archive.org/details/eliza\_1966\_mad\_slip\_src
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